This report is from the running model.

CHURN MODEL · LOGREG
DRIFT · LAST 151 OF 200

feature	train → recent · z
Tenure	35.3 → 58.0 mo · z 13.3
Monthly	$70 → $109 · z 16.5
Total	$2,483 → $6,891 · z 28.4
Senior	0.53 → 0.46 · z -1.7
Contract	max shift 0.06 · ok

served	151
p50 / p99 ms	1.35 / 1.81
threshold	|z| > 3

DRIFT · Tenure, Monthly, Total
GET /drift · window 200

The inputs moved. The model kept answering.

Who notices before a customer does?

mean tenure, random traffic

32.3 mo

↓

mean tenure, shifted traffic

58.0 mo

Maximus Friedrich Baluyot

Model monitoring API

Portfolio · ML Engineer

Every request counted, timed and checked for drift

```
max@baluyot ~ $ curl -s -XPOST
':8115/simulate?n=100&shifted=true&s
eed=2' >/dev/null && curl -s
:8115/drift
```



1 Training statistics stored in the bundle; traffi
`pandas`

2 Features
`fastapi`

3 The churn champion pipeline from ../churn_pr…
`fastapi · prometheus_client · /metr…`

4 FastAPI + one HTML status page on common
`FastAPI`

5 Dashboard
`HTML + ui.css`

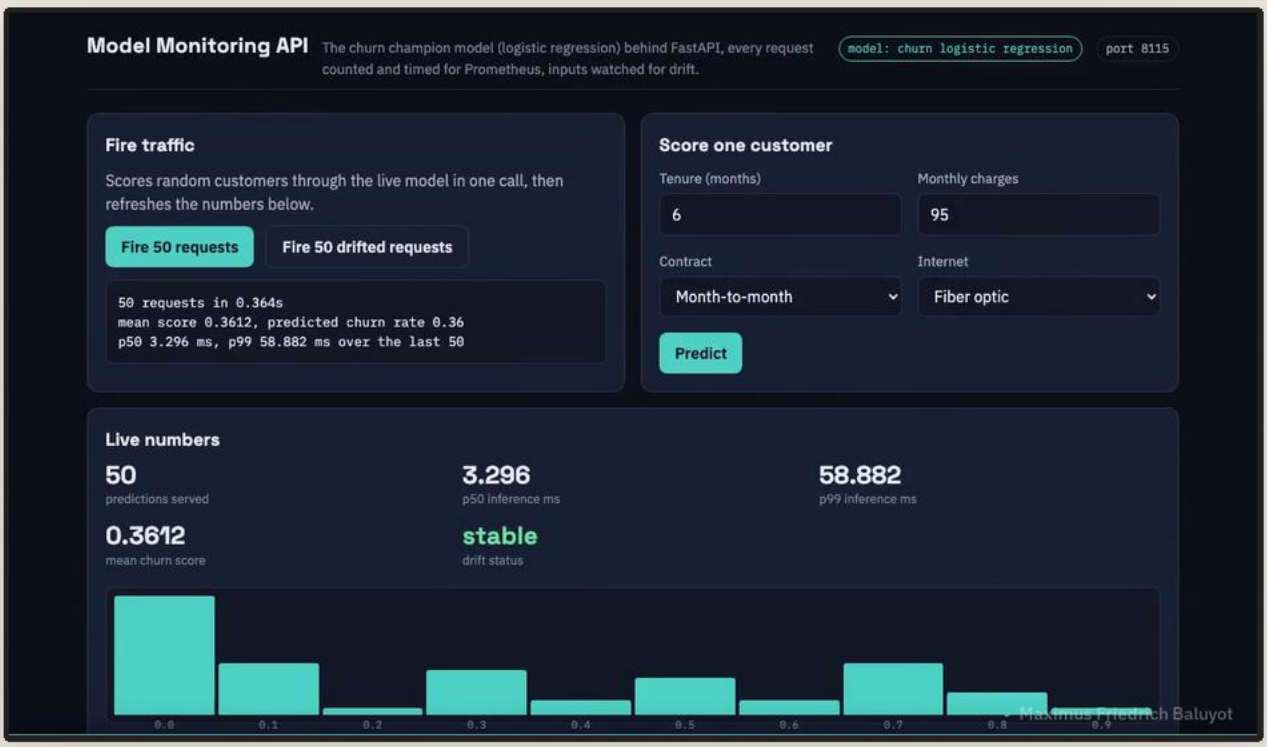
Standard error of the mean, so sample size counts

Maximus Friedrich Baluyot

GET /drift after 100 shifted requests

feature	train mean	recent mean	z	flag
Tenure	**35.34**	**58.04**	**13.3**	**drift**
MonthlyCharges	70.10	109.05	16.5	drift
TotalCharges	2,483	6,891	28.4	drift
SeniorCitizen	0.53	0.46	-1.7	ok

Fifty requests: served, p50, p99, drift status



z 13.3

Tenure drift · 100 shifted requests · flagged above 3

one limit

Traffic is simulated. The window resets on restart.

Real model, real metrics. And it ships.

Maximus Friedrich Baluyot

Try it: Fire 50 drifted requests, twice.

```
max@baluyot ~ $ git clone
https://github.com/Keshigami/
model_monitoring_api.git
max@baluyot ~ $ just
model_monitoring_api-setup
max@baluyot ~ $ just
model_monitoring_api-up
max@baluyot ~ $
```

github.com/Keshigami/model_monitoring_api

Maximus Friedrich Baluyot

AI & ML Specialist · Applied AI · ML Engineering · Automation